

Analysis of Flight Risk, Delays, and Cost Efficiency

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ABSTRACT

Background: This project addresses declining consumer trust in air travel by analyzing flight data to develop risk profiles for airlines and routes, focusing on costs, delay probabilities, severity, and reliability to improve safety, efficiency, and passenger satisfaction.

Methods: This project preprocesses and analyzes a flight dataset using cleaning techniques like forward filling and one-hot encoding, establishes baselines with static models, and applies advanced methods like SVM and MLP for classification and regression tasks, evaluated through metrics such as accuracy, precision, RMSE, and R^2 , ensuring robust predictive insights into flight performance and cost efficiency.

Results: SVM models outperformed MLP in predicting delay severity, effectively handling imbalanced labels and achieving higher accuracy , recall, and precision (88%) for delayed flights. For cost-per-mile regression, MLP regressors achieved strong accuracy ($R^2 \sim 0.87$) and low RMSE.

Conclusions: These findings underscore the potential of machine learning in enhancing airline operations and passenger satisfaction by providing actionable insights into delay patterns and cost efficiency.

BACKGROUND

Previous studies, such as the HFACS analysis of aviation accidents and the MCAS crash examination, focus on post hoc evaluations of systemic failures, while our project shifts toward proactive risk profiling using machine learning models like SVM and MLP to predict flight delay severity and cost-per-mile. By integrating data-driven insights, our approach complements existing research by identifying high-risk routes and inefficiencies, helping to improve airline operations and passenger outcomes.

OBJECTIVE

The objective of this project is to analyze flight data to create risk profiles for airlines and routes, focusing on costs, delay probabilities, severity, and reliability, to enhance safety, operational efficiency, and passenger satisfaction within the commercial airline industry.

METHODS

Study design / Inclusion

- Developed a methodology to predict flight delay severity (classification) and cost-per-mile (regression) using machine learning models like SVM and MLP.
- Established baseline models with static thresholds for classification and linear regression for cost prediction to benchmark advanced techniques.
- Incorporated advanced model features such as dropout layers and early stopping to ensure robust performance and prevent overfitting.

Data Collection

- Utilized flight data spanning six years (2018–2024) with features like departure delays, arrival delays, distance, elapsed times, and categorical variables (airline, origin, destination).
- Preprocessed data by cleaning missing values via forward filling, removing canceled and diverted flights, and encoding categorical variables using one-hot encoding.
- Engineered additional features, including CostPerMile and Delay Severity.

Statistical analysis

- Evaluated classification models using metrics such as accuracy, precision, recall, F1-score, and confusion matrices to measure predictive performance.
- Assessed regression models through RMSE, MAE, R^2 , and validation loss to ensure accuracy and generalization.
- Visualized model performance through loss curves, scatter plots, and post-hoc error analysis, uncovering biases and areas for improvement.

RESULTS

Figure 1. Cost per Mile Prediction Plots

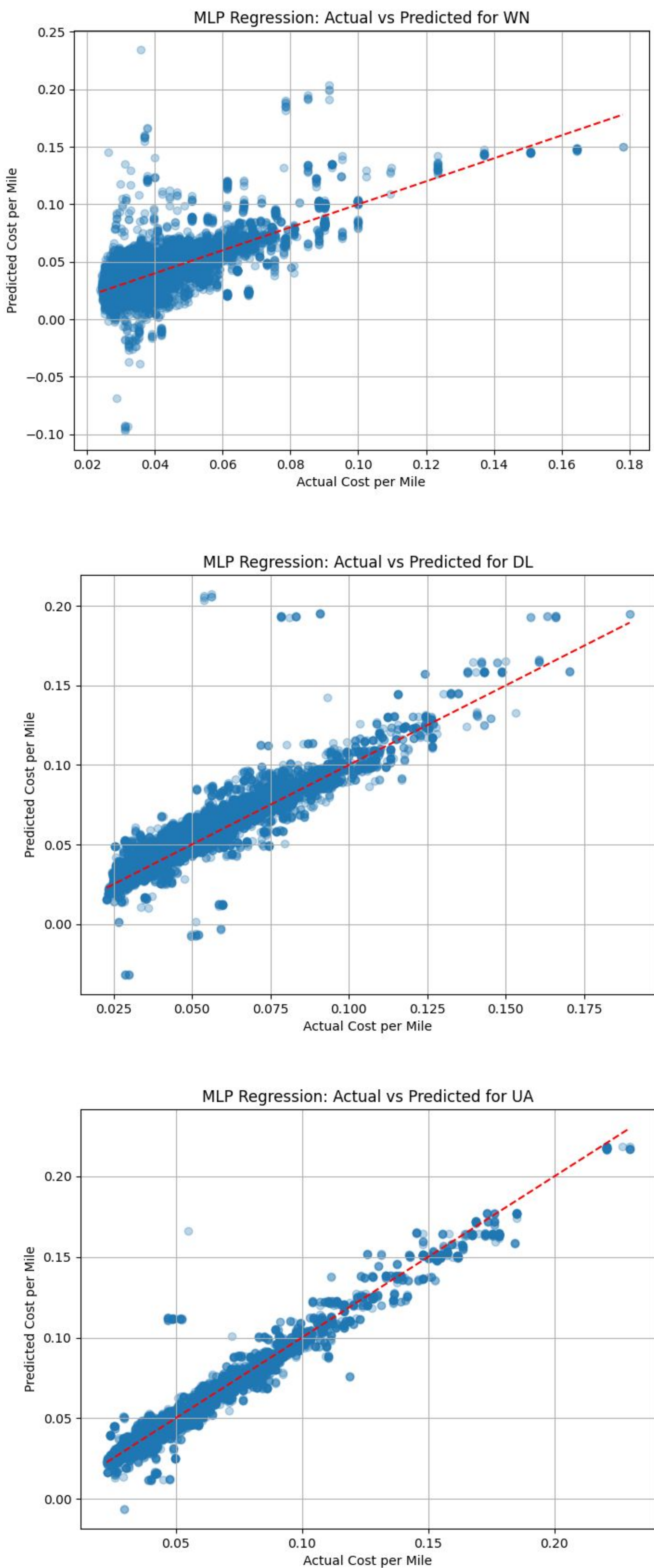


Table 1. Delay Prediction Results

Aspect	SVM	MLP
Accuracy	88%	60%
Recall (Delayed)	High (86.6%)	Average (50%)
Precision (Delayed)	Strong (78.6%)	Low (30%)

- The SVM model achieves high accuracy and recall for delays but overpredicts them, with lower precision for delay classes.
- The MLP model fails to flag delayed flights, achieving, making SVM superior for delay prediction.

Table 2. Cost Per Mile Prediction Results

Airline	RMSE	R2 Score
Southwest	0.0191	0.8756
Delta	0.0186	0.8879
United	0.0178	0.8912

CONCLUSIONS

- Machine learning models like SVM and MLP offer promising approaches for predicting flight delay severity and cost-per-mile, providing valuable insights to enhance airline efficiency and passenger satisfaction.
- Continued improvements in feature engineering and addressing data limitations, such as class imbalances and outliers, can further refine predictive accuracy and operational reliability.